Data Description, Dictionary, and Exploration Report

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1. Data Description Report

1.1 Data Requirements

Required data characteristics:

- A labelled target variable indicating diabetes status (no diabetes, prediabetes, diabetes)
- Health measurements: BMI, blood pressure status, cholesterol status
- Lifestyle indicators: physical activity, dietary habits, smoking, alcohol consumption
- Demographic features: age, sex, income level, education level
- Sufficient sample size for machine learning
- Publicly available data with no access restrictions

1.2 Assumptions and Limitations

- The dataset provided is accurate and correctly labeled
- It is representative of the target population
- Features included are sufficient for predicting diabetes risk
- No major data collection errors exist
- User inputs during deployment follow the same format as training data

1.3 Dataset Overview Results

Shape: 253,680 rows × 22 columns. Memory usage: approximately 42.6 MB.

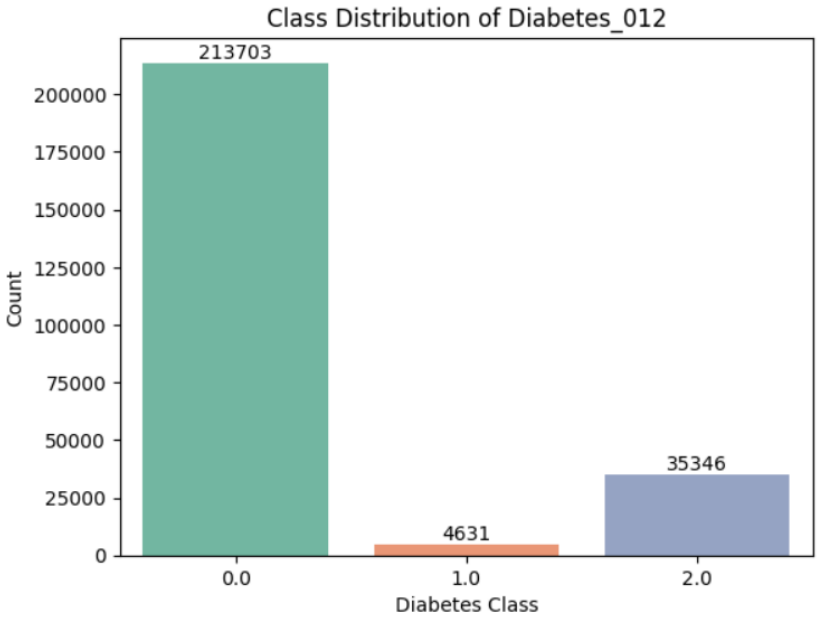
Data types: All 22 columns stored as float64. Binary columns encoded as 0/1.

Feature breakdown: 1 target variable + 14 binary features + 4 ordinal features + 3 continuous features.

Target variable distribution:

Class	Label	Count	Percentage	Ratio vs Class 0
0	No diabetes	213,703	84.2%	1 : 1
1	Prediabetes	4,631	1.8%	46 : 1
2	Diabetes	35,346	13.9%	6 : 1

The dataset is severely imbalanced. Prediabetes represents only 1.8% of records with a 46:1 ratio against the majority class. This imbalance must be addressed during modelling.



2. Data Dictionary

The data dictionary documents the schema of the dataset including feature name, description, data type, value range, semantic context of each code, and measurement level. This artifact is essential for data governance, reproducibility, and team collaboration.

Feature	Description	Type	Range	Codes	Level
Diabetes_012	Diabetes status	Categorical	0–2	0=No, 1=Pre, 2=Yes	Nominal
HighBP	High blood pressure	Binary	0–1	0=No, 1=Yes	Nominal
HighChol	High cholesterol	Binary	0–1	0=No, 1=Yes	Nominal
CholCheck	Chol. check in 5 yrs	Binary	0–1	0=No, 1=Yes	Nominal
BMI	Body Mass Index	Continuous	12–98	Numeric value	Ratio
Smoker	Smoked 100+ cigs	Binary	0–1	0=No, 1=Yes	Nominal
Stroke	Ever had stroke	Binary	0–1	0=No, 1=Yes	Nominal
HeartDiseaseorAttack	Heart disease/attack	Binary	0–1	0=No, 1=Yes	Nominal
PhysActivity	Active in 30 days	Binary	0–1	0=No, 1=Yes	Nominal
Fruits	Fruit 1+/day	Binary	0–1	0=No, 1=Yes	Nominal
Veggies	Veg 1+/day	Binary	0–1	0=No, 1=Yes	Nominal
HvyAlcoholConsump	Heavy drinker	Binary	0–1	0=No, 1=Yes	Nominal
AnyHealthcare	Has coverage	Binary	0–1	0=No, 1=Yes	Nominal
NoDocbcCost	No doc due to cost	Binary	0–1	0=No, 1=Yes	Nominal
GenHlth	General health	Ordinal	1–5	1=Exc...5=Poor	Ordinal
MentHlth	Bad mental days	Continuous	0–30	Days in month	Ratio
PhysHlth	Bad physical days	Continuous	0–30	Days in month	Ratio
DiffWalk	Walking difficulty	Binary	0–1	0=No, 1=Yes	Nominal
Sex	Gender	Binary	0–1	0=F, 1=M	Nominal
Age	Age category	Ordinal	1–13	1=18-24...13=80+	Ordinal
Education	Education level	Ordinal	1–6	1=None...6=College	Ordinal
Income	Income level	Ordinal	1–8	1=<$10k...8=>$75k	Ordinal

3. Data Exploration Report

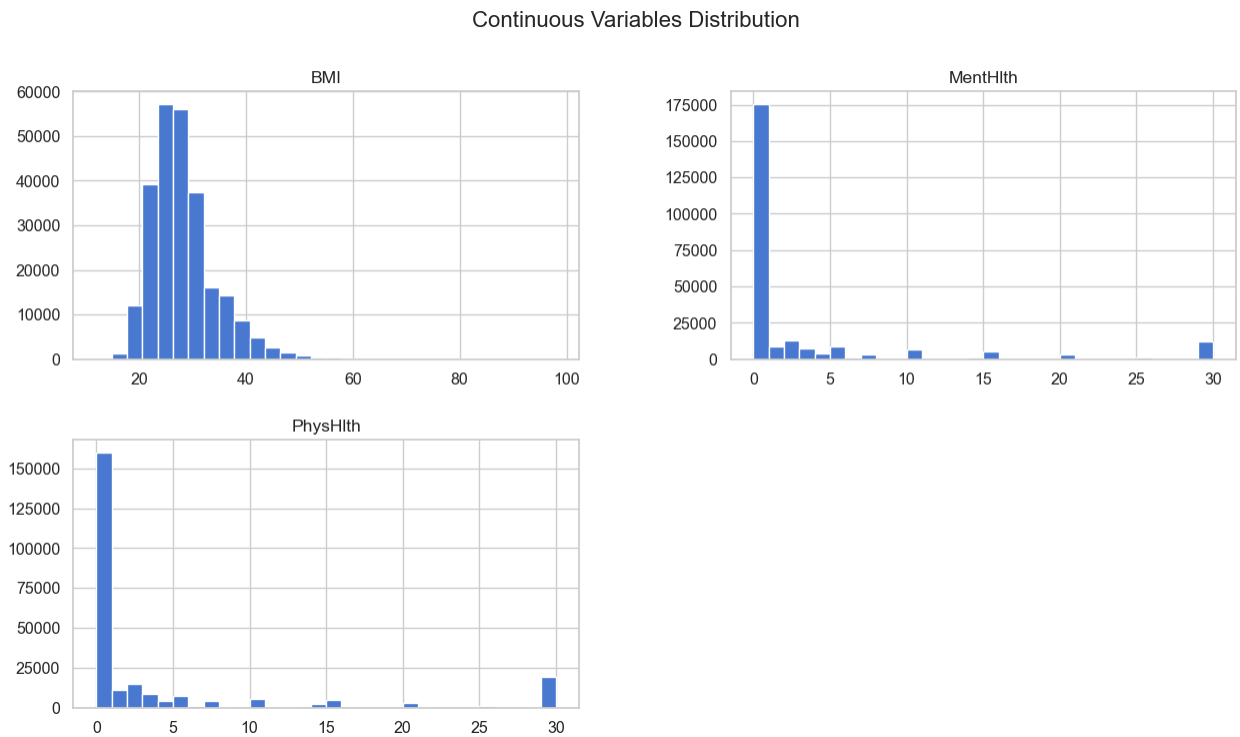
This section presents the results of exploratory data analysis conducted on the diabetes dataset. The analysis covers univariate distributions, bivariate relationships with the target variable, correlation analysis, and outlier detection. Each subsection states the goal, presents results, and draws implications for data preparation and modelling.

3.1 Univariate Analysis

3.1.1 Continuous Features (BMI, MentHlth, PhysHlth)

Goal: Understand the distribution shape, central tendency, and spread of each continuous feature. This helps identify distributions, class imbalance, skewness, unusual values, and dominant categories before modeling.

Results:



Feature	Mean	Median	Std Dev	Min	Max	Skewness	Kurtosis
BMI	28.38	27.0	6.61	12	98	2.12	11.00
MentHlth	3.18	0.0	7.41	0	30	2.72	6.44
PhysHlth	4.24	0.0	8.72	0	30	2.21	3.50

Key findings:

- BMI is approximately normally distributed but right-skewed (skew=2.12). The mean of 28.38 falls in the overweight range. Values above 50 are extreme outliers and BMI=98 may be a data entry error.
- MentHlth and PhysHlth are zero-inflated: most respondents report 0 bad days (median=0), with a long right tail up to 30. These distributions are strongly right-skewed.

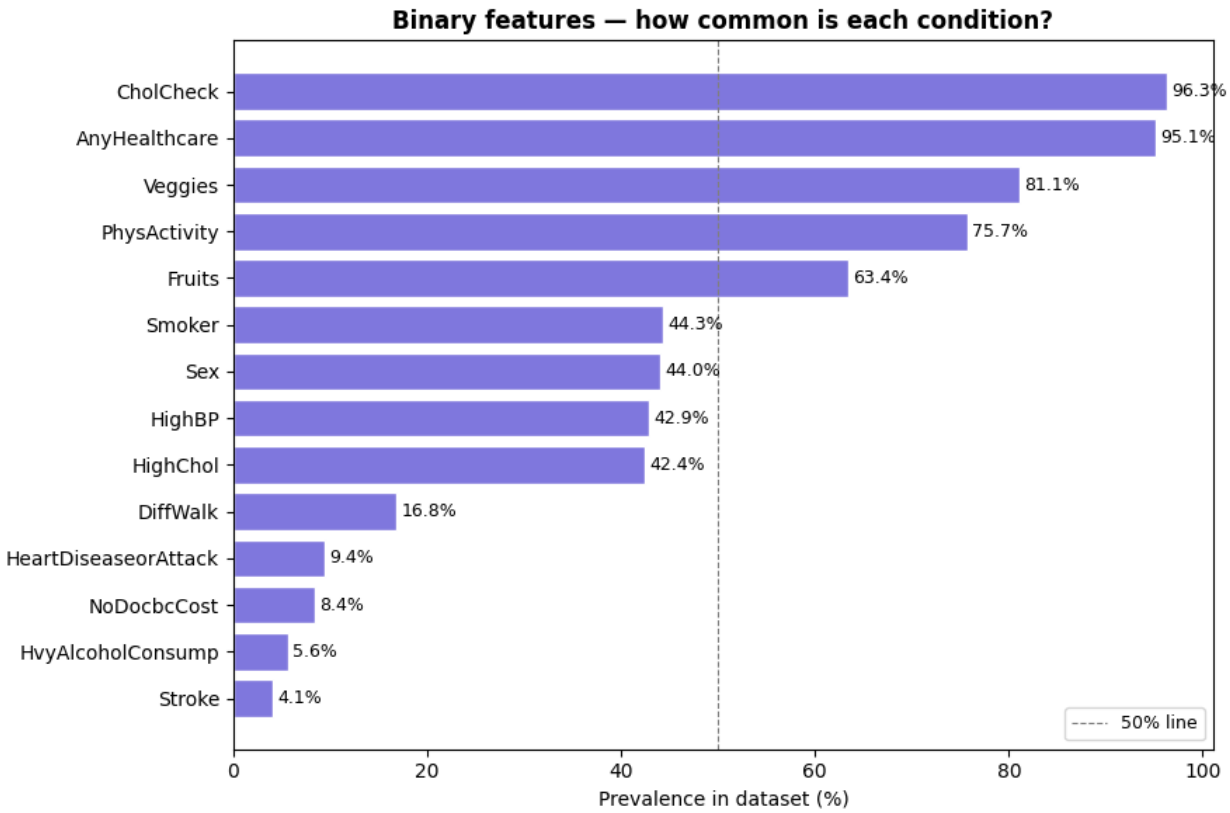
- All three features fail normality tests, indicating non-parametric methods should be used for group comparisons.

Implication for preparation: Consider capping BMI at the 98th percentile (45) to limit extreme values. MentHlth and PhysHlth outliers are valid chronic illness reports and should be retained.

3.1.2 Binary Features

Goal: Assess the prevalence (% answering Yes) of each binary condition in the survey population.

Results:

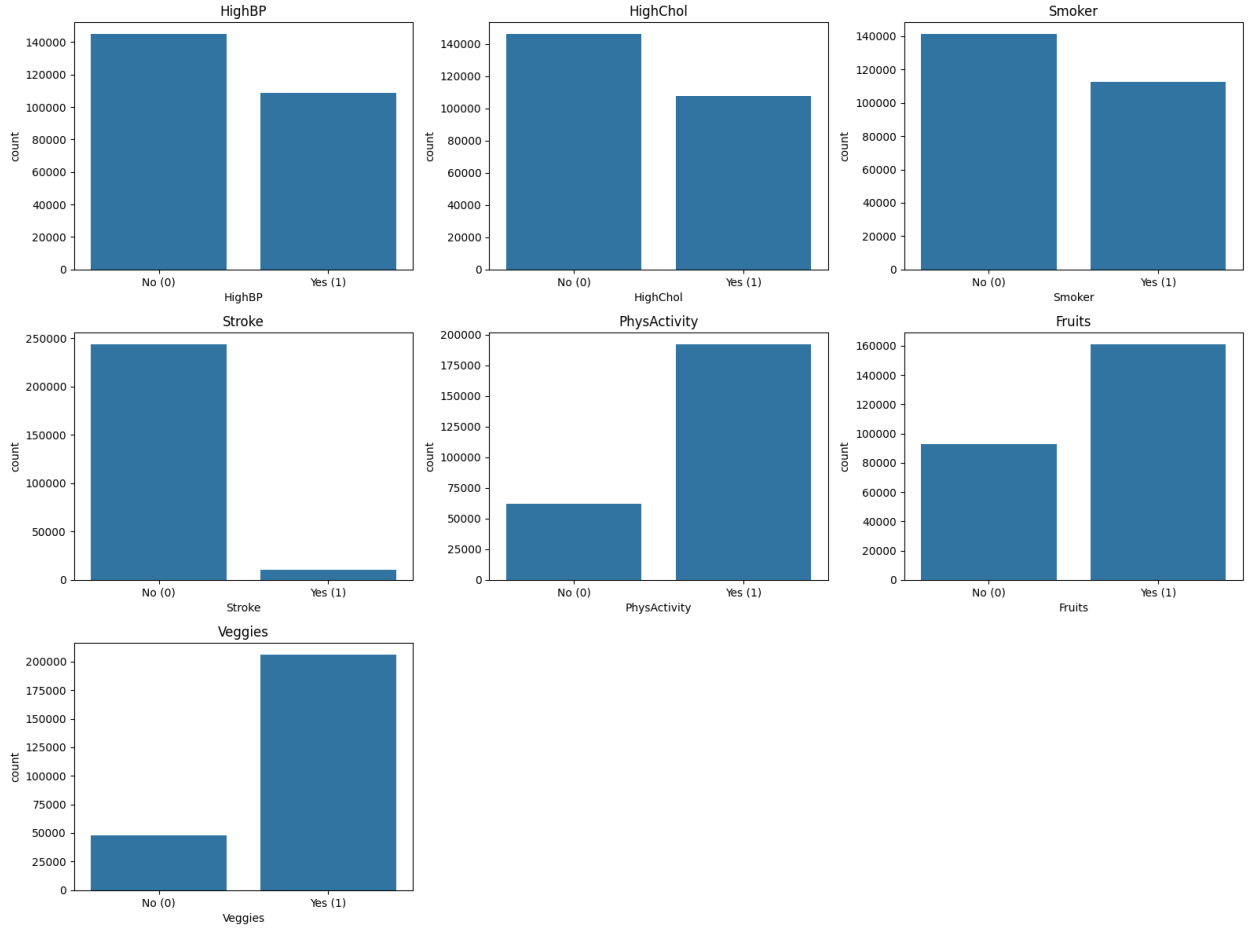


Feature	Prevalence (% = 1)	Observation
CholCheck	96.3%	Near-constant — almost everyone had a check
AnyHealthcare	95.1%	Near-constant — almost everyone has coverage
Veggies	81.1%	Most people eat vegetables daily
PhysActivity	75.7%	Most people exercise
Fruits	63.4%	Majority eat fruit daily
Smoker	44.3%	Nearly half are lifetime smokers
HighBP	42.9%	Common condition
HighChol	42.4%	Common condition

Sex (Male)	44.0%	Roughly balanced gender split
DiffWalk	16.8%	Minority has walking difficulty
HeartDisease	9.4%	Relatively uncommon
NoDocbcCost	8.4%	Small % skipped doctor due to cost
HvyAlcohol	5.6%	Low prevalence
Stroke	4.1%	Rare condition

Key finding: CholCheck (96.3%) and AnyHealthcare (95.1%) have near-zero variance. When a feature is almost always the same value, it cannot help distinguish diabetic from non-diabetic patients.

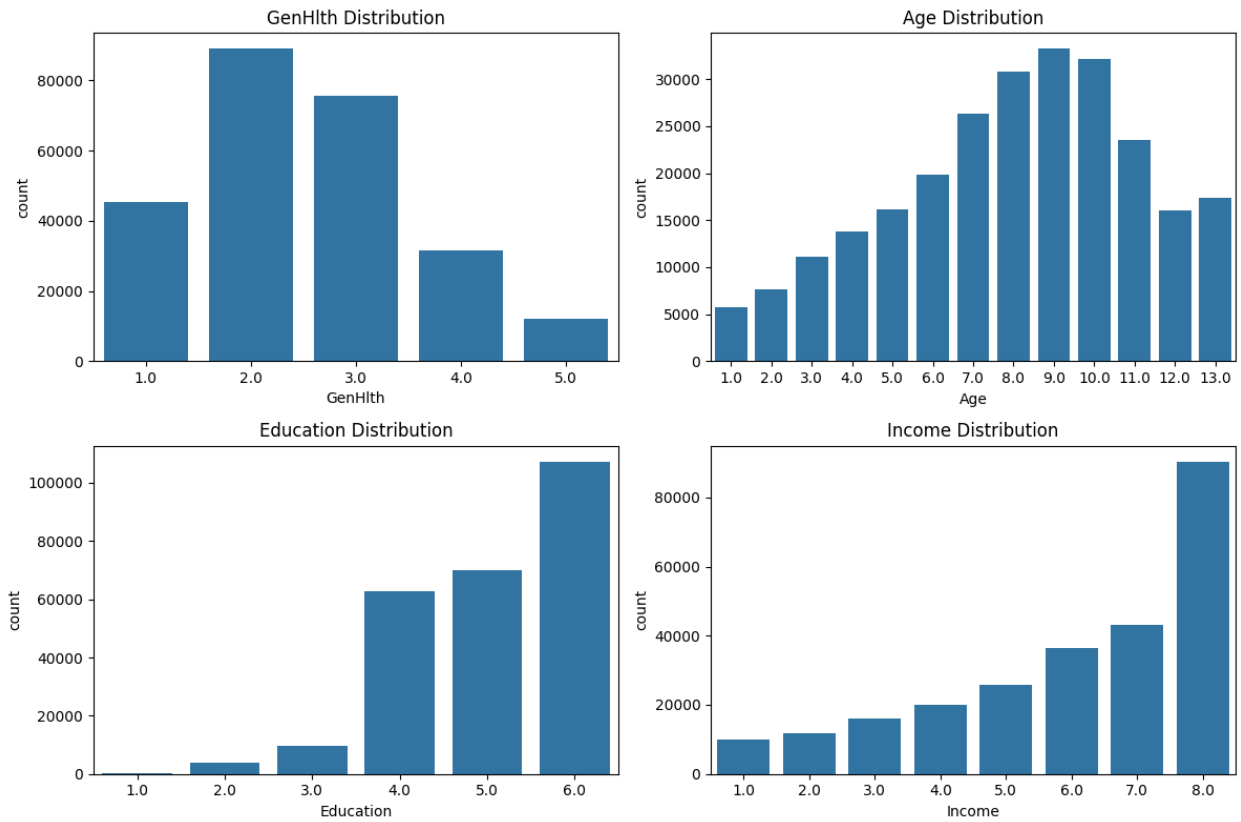
Implication: Consider dropping CholCheck and AnyHealthcare before modelling due to insufficient variance.



3.1.3 Ordinal Features

Goal: Understand the distribution of ordinal categories across the population.

Results:



Key findings:

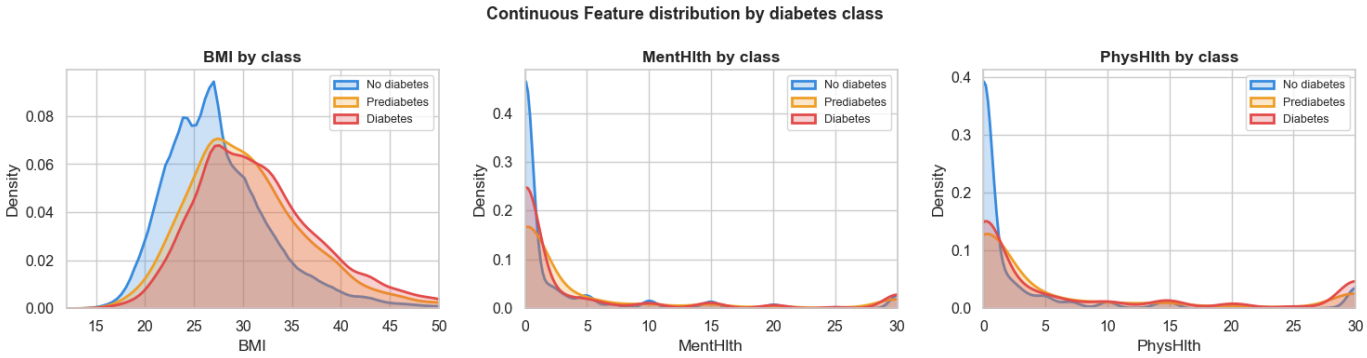
- GenHlth: peaks at 2 (Very good) and 3 (Good). Most respondents rate their health positively.
- Age: the dataset is skewed toward older adults, peaking in groups 8–11 (55–74 years). Slight bias toward middle-aged / older population
- Education: heavily concentrated at levels 5–6 (Some college and College graduate). Few respondents have low education.
- Income: skewed toward higher brackets (7–8), Could introduce bias in predictions.

3.2 Bivariate Analysis

Goal: Determine how each feature relates to the target variable (Diabetes_012). This section answers: does having a specific condition or behaviour change the likelihood of diabetes?

3.2.1 Continuous Features vs Diabetes Rate

Results:

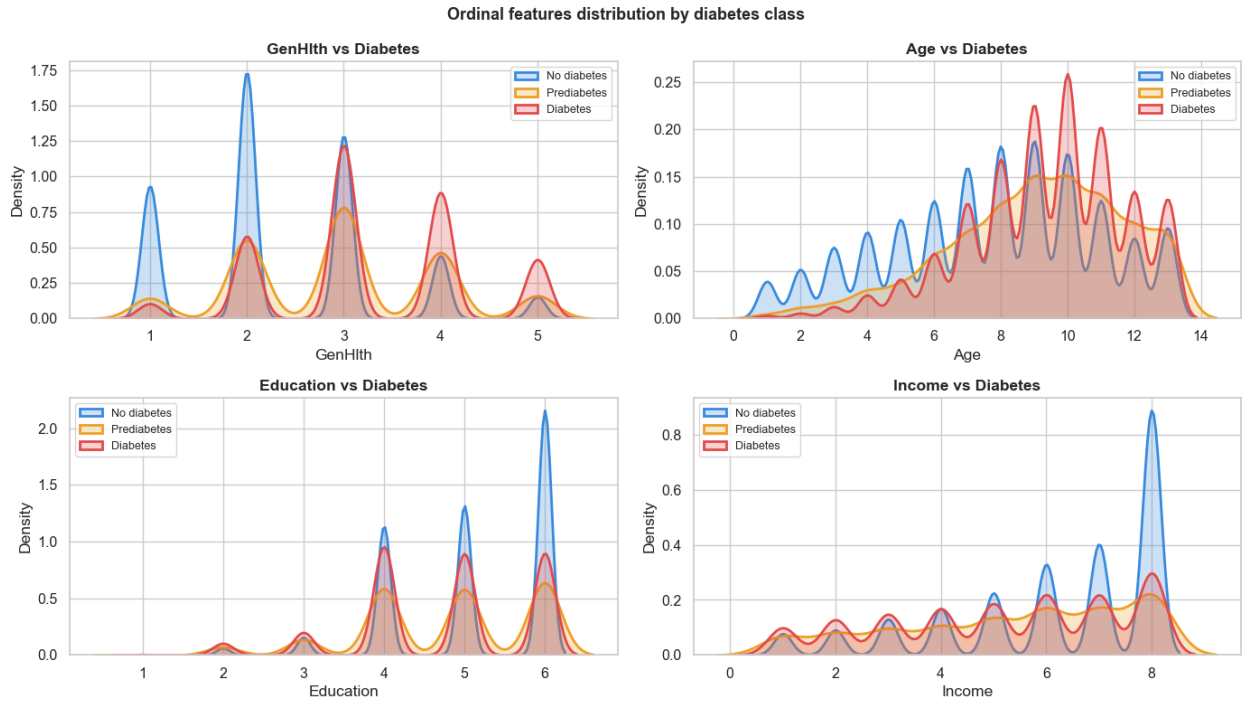


Key findings:

- BMI: Higher BMI is strongly associated with diabetes; diabetic and prediabetic groups are shifted toward higher BMI values compared to non-diabetic individuals.
- MentHlth: Most individuals report 0 unhealthy days; however, diabetic individuals show a slightly higher number of poor mental health days (weak association).
- PhysHlth: Diabetic individuals report more physical unhealthy days, showing a clearer separation than mental health (moderate association).
- Overall: There is noticeable overlap between classes across all features, indicating that no single feature alone can accurately predict diabetes.

3.2.2 Ordinal Features vs Diabetes Rate

Results:

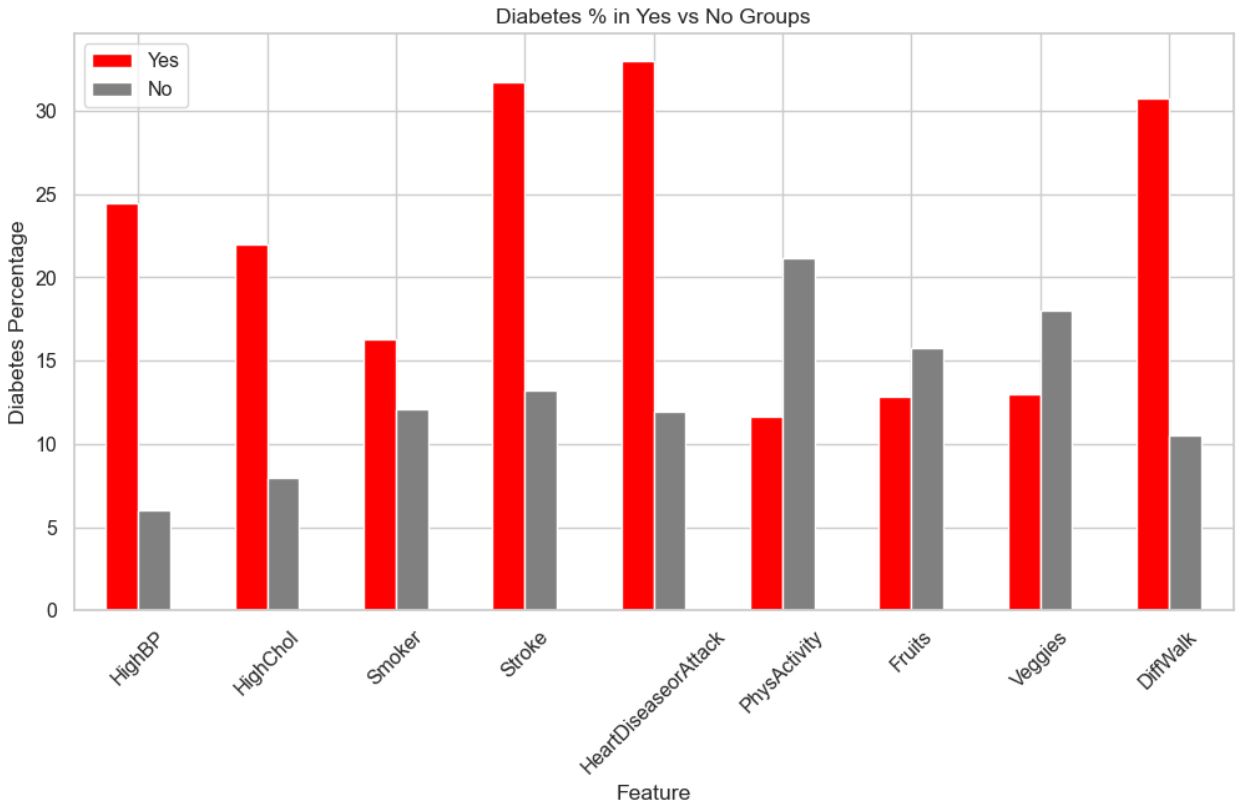


Key findings:

- GenHlth: Diabetic individuals are concentrated in poorer health categories (3–5), while non-diabetic individuals are mostly in better health (1–2), indicating a strong relationship.
- Age: Diabetes prevalence increases with age; diabetic individuals are more concentrated in higher age groups (8–12).
- Education: Higher education levels are more common in non-diabetic individuals, while diabetic individuals are relatively more spread across lower education levels (weak to moderate relationship).
- Income: Lower income groups show a higher proportion of diabetes, while non-diabetic individuals are more concentrated in higher income categories (moderate relationship).
- Overall: Clear trends exist, especially for GenHlth and Age, but class overlap remains across all features.

3.2.3 Binary Features vs Diabetes Rate

Method: For each binary feature, we calculated the diabetes rate (% with class 2) among people with the condition (=1) versus without it (=0). The difference shows whether the condition is a risk factor (positive difference) or protective (negative difference).



Key findings:

- HeartDiseaseorAttack shows the largest gap: 33.0% vs 12.0%. People with heart disease have nearly 3x the diabetes rate.
- DiffWalk (+20.2%), Stroke (+18.6%), and HighBP (+18.4%) are also very strong risk indicators.
- PhysActivity is the strongest protective factor: exercisers have 11.6% diabetes rate versus 21.1% for non-exercisers (−9.5% difference). This proves physical activity is genuinely associated with lower diabetes risk in this dataset.
- Veggies (−5.0%) and HvyAlcoholConsump (−8.6%) are also protective. The alcohol finding may reflect survivorship bias rather than a genuine protective effect.
- Sex and AnyHealthcare show negligible differences (<3%), confirming they are weak predictors.

3.3 Correlation Analysis

Goal: Measure linear relationships between features and the target, and identify multicollinearity between feature pairs.

3.3.1 Feature Correlations with Target

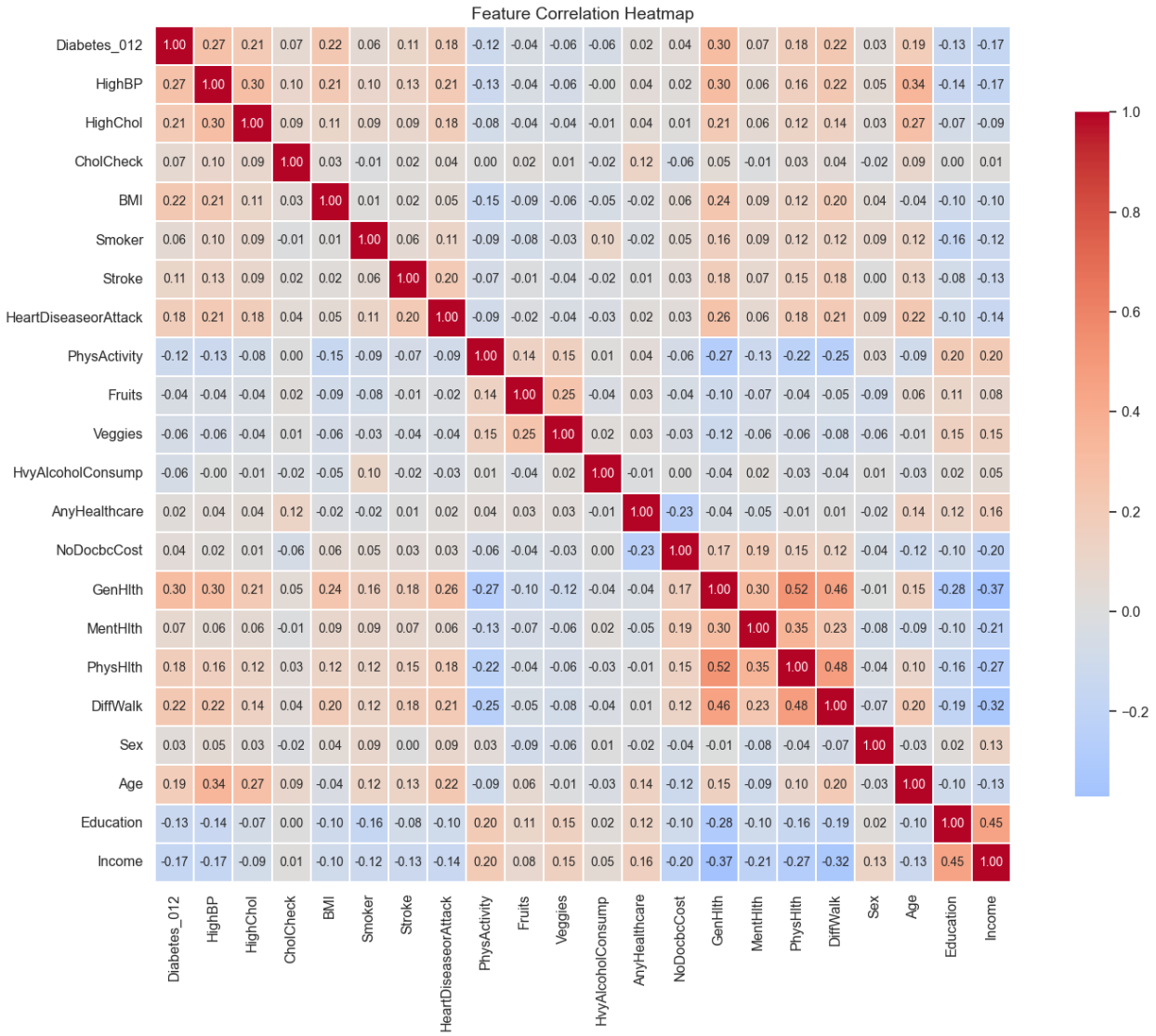
Method: Pearson correlation coefficient computed for each feature against Diabetes_012.

Feature	Correlation	Direction	Strength
GenHlth	+0.303	MORE diabetes	Strong
HighBP	+0.272	MORE diabetes	Moderate-Strong
BMI	+0.224	MORE diabetes	Moderate
DiffWalk	+0.224	MORE diabetes	Moderate
HighChol	+0.209	MORE diabetes	Moderate
Age	+0.185	MORE diabetes	Moderate
HeartDisease	+0.180	MORE diabetes	Moderate
PhysHlth	+0.176	MORE diabetes	Moderate
Stroke	+0.107	MORE diabetes	Weak
MentHlth	+0.074	MORE diabetes	Weak
PhysActivity	−0.122	LESS diabetes	Moderate
Education	−0.131	LESS diabetes	Moderate
Income	−0.171	LESS diabetes	Moderate

Key finding: GenHlth has the strongest correlation (+0.303) with diabetes. Income (−0.171) is the strongest protective correlation. Features like Sex (+0.031) and AnyHealthcare (+0.015) have near-zero correlations, confirming their weak predictive value.

3.3.2 Multicollinearity (Feature-Feature Correlations)

Method: Identified feature pairs with |correlation| > 0.30, which indicates shared information.



Feature A	Feature B	Correlation	Implication
GenHlth	PhysHlth	0.524	Strong overlap — both measure health status
PhysHlth	DiffWalk	0.478	Physical impairment cluster
GenHlth	DiffWalk	0.457	Health-mobility cluster
Education	Income	0.449	Socioeconomic cluster
GenHlth	Income	−0.370	Lower income = worse health
MentHlth	PhysHlth	0.354	Mental-physical health link
HighBP	Age	0.344	BP increases with age
DiffWalk	Income	−0.320	Lower income = more mobility issues
GenHlth	MentHlth	0.302	General health includes mental component
HighBP	GenHlth	0.301	BP linked to poor general health

Key finding: GenHlth, PhysHlth, and DiffWalk form a tightly correlated cluster (r=0.45–0.52). Using all three in a linear model introduces multicollinearity. Education and Income share 45% of their signal. For linear models, consider dropping one from each correlated pair or using regularization (L1/L2). Tree-based models handle multicollinearity natively.

3.3.3 Exploration Summary

- **Strong features (definitely keep):** GenHlth, HighBP, BMI, DiffWalk, HighChol, Age, HeartDiseaseorAttack
- **Moderate features (keep):** PhysHlth, Income, Education, PhysActivity, Stroke
- **Weak features (consider dropping):** MentHlth, Smoker, CholCheck, NoDocbcCost, Sex, AnyHealthcare
- **Near-constant (drop):** CholCheck (96.3% same value), AnyHealthcare (95.1% same value)
- **Correlated pairs to watch:** GenHlth↔PhysHlth (0.52), PhysHlth↔DiffWalk (0.48), Education↔Income (0.45)